An Euler Product over the Inert Primes of $\mathbb{Q}(\sqrt{3})$

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Abstract

We prove the prime-product conjecture in OEIS A003630. The partial product is exactly 9/8 times a quotient of two partial Dirichlet L-products. Their convergence in increasing prime order, together with two elementary L-value integrals, gives the conjectured limit $\sqrt{3}/2$.

1 The product and its character identity

The inert rational primes of $\mathbb{Q}(\sqrt{3})$ are exactly the primes congruent to 5 or 7 modulo 12. This classification is already established in A003630 [1], including Jianing Song's 2018 comment. The separate conjecture of Dimitris Valianatos, dated March 26, 2017, concerns the following product.

Theorem 1. *With factors taken in increasing order of the primes,*

$$\prod_{\substack{p \text{ prime} \\ p\equiv 5,7 \pmod{12}}} \begin{cases} (p-1)/(p+1), & p\equiv 1 \pmod 4, \\ (p+1)/(p-1), & p\equiv 3 \pmod 4 \end{cases} = \frac{\sqrt{3}}{2}.$$

Define the two real nonprincipal characters

$$\chi_4(n) = \begin{cases} 0, & 2 \mid n, \\ 1, & n\equiv 1 \pmod 4, \\ -1, & n\equiv 3 \pmod 4, \end{cases} \qquad \chi_{-3}(n) = \begin{cases} 0, & 3 \mid n, \\ 1, & n\equiv 1 \pmod 3, \\ -1, & n\equiv 2 \pmod 3. \end{cases}$$

For $X \geq 3$, put

$$L_X(1,\chi) = \prod_{p\leq X} \left(1 - \frac{\chi(p)}{p}\right)^{-1}, \qquad P(X) = \prod_{\substack{p\leq X \\ p\equiv 5,7 \pmod{12}}} \left(\frac{p-1}{p+1}\right)^{\chi_4(p)}.$$

We first work with finite products. At every prime $p > 3$ the local factor of $L_X(1,\chi_{-3})/L_X(1,\chi_4)$ is

$$\frac{1-\chi_4(p)/p}{1-\chi_{-3}(p)/p}.$$

It is one for $p \equiv 1, 11 \pmod{12}$, and is respectively $(p-1)/(p+1)$ and its reciprocal for $p \equiv 5, 7 \pmod{12}$. The two exceptional factors at $p = 2, 3$ are $2/3$ and $4/3$. Thus the exact identity is

$$P(X) = \frac{9}{8}\frac{L_X(1,\chi_{-3})}{L_X(1,\chi_4)} \qquad (X \geq 3). \tag{1}$$

2 Convergence and evaluation

We justify the ordinary prime-ordered limit, rather than only an Abelian limit with exponent $s \downarrow 1$.

Lemma 1. *For either character $\chi = \chi_4, \chi_{-3}$,*

$$\lim_{X\to\infty} L_X(1,\chi) = L(1,\chi) = \sum_{n\ge 1} \frac{\chi(n)}{n}.$$

Proof. We use the established prime-number theorem estimate

$$\theta(x;q,a) = \frac{x}{\varphi(q)} + O_q(x/\log x), \qquad (a,q) = 1,$$

for $q = 3, 4$, supplied, for example, by Theorem 1.2 of Bennett–Martin–O'Bryant–Rechnitzer [2]. Since a nonprincipal character sums to zero over its unit classes,

$$\Theta_\chi(x) := \sum_{p\le x} \chi(p)\log p = O(x/\log x).$$

Partial summation with weight $1/(t\log t)$ gives

$$\sum_{p\le X} \frac{\chi(p)}{p} = \frac{\Theta_\chi(X)}{X\log X} + \int_2^X \Theta_\chi(t)\frac{\log t + 1}{t^2(\log t)^2}\,dt.$$

The endpoint tends to zero and the integral converges, its large-t integrand being $O(1/(t(\log t)^2))$. Hence the prime sum converges. The higher terms in

$$-\log(1-\chi(p)/p) = \frac{\chi(p)}{p} + \sum_{k\ge 2} \frac{\chi(p)^k}{kp^k}$$

are absolutely summable over primes. The logarithms, and therefore the positive partial Euler products, converge to a nonzero limit. Abel's theorem applied to the convergent prime sum, with weights $p^{-(s-1)}$, identifies the logarithmic limit as $\lim_{s\downarrow 1} \log L(s,\chi)$: the higher-power terms pass to the limit by absolute domination. Finally, bounded partial sums of the periodic nonprincipal character give continuity of its Dirichlet series from the right at $s = 1$. The Euler product for $s > 1$ therefore identifies the limit with $L(1,\chi)$. □

Both required values have elementary integral evaluations. Pairing terms of the defining character series and summing their geometric series under the integral gives

$$L(1,\chi_4) = \sum_{j\ge 0}\left(\frac{1}{4j+1} - \frac{1}{4j+3}\right) = \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4},$$

$$L(1,\chi_{-3}) = \sum_{j\ge 0}\left(\frac{1}{3j+1} - \frac{1}{3j+2}\right) = \int_0^1 \frac{dt}{1+t+t^2} = \frac{\pi}{3\sqrt{3}}.$$

The paired integrands are nonnegative, justifying these exchanges by monotone convergence. Taking the limit in (1) and using Lemma 1 now proves Theorem 1:

$$\lim_{X\to\infty} P(X) = \frac{9}{8}\frac{\pi/(3\sqrt{3})}{\pi/4} = \frac{\sqrt{3}}{2}.$$

The accompanying exact verifier checks all 1000 published prime terms and the local quadratic-residue classification, the rational identity (1) at every prime cutoff through 10000, and the exceptional factors separately. Numerical partial products are supplementary controls; convergence follows from the analytic argument and the cited prime-number theorem estimate.

References

[1] The OEIS Foundation Inc., A003630, accessed October 27, 2026.

[2] M. A. Bennett, G. Martin, K. O'Bryant and A. Rechnitzer, *Explicit bounds for primes in arithmetic progressions*, Theorem 1.2, arXiv:1802.00085, version 3, 2018.